Q1. JavaScript Coding Practice

1. Check Whether a Number is Even or Odd

Problem Statement

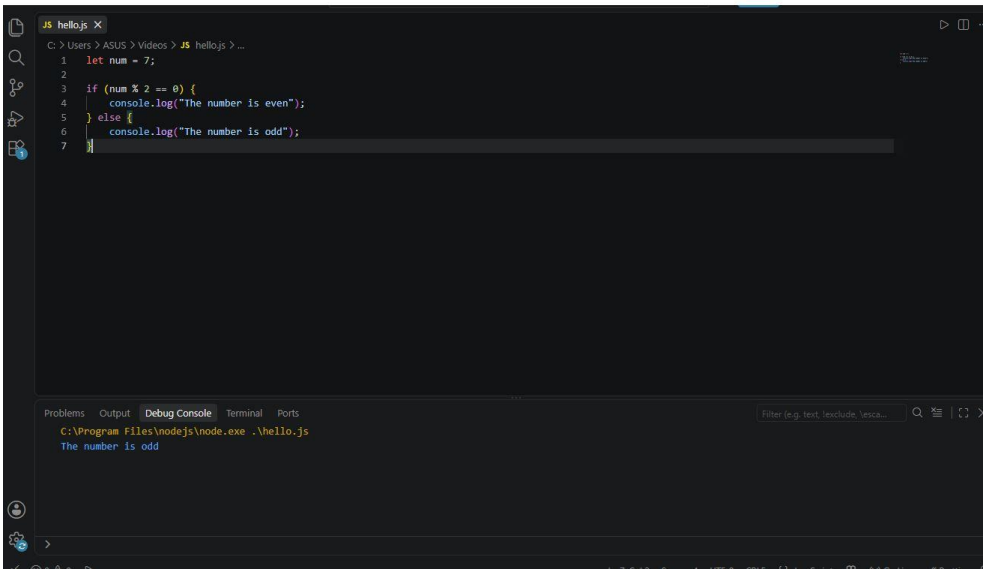
Write a JavaScript program to check whether a given number is even or odd.

Source Code

```
let num = 7;

if (num % 2 == 0) {
    console.log("The number is even");
} else {
    console.log("The number is odd");
}
```

Output



Output: The number is odd.

Explanation

The program uses the remainder operator (%) to divide the number by 2. If the remainder is 0, the number is even. Otherwise, it is odd.

2. Find the Largest of Three Numbers

Problem Statement

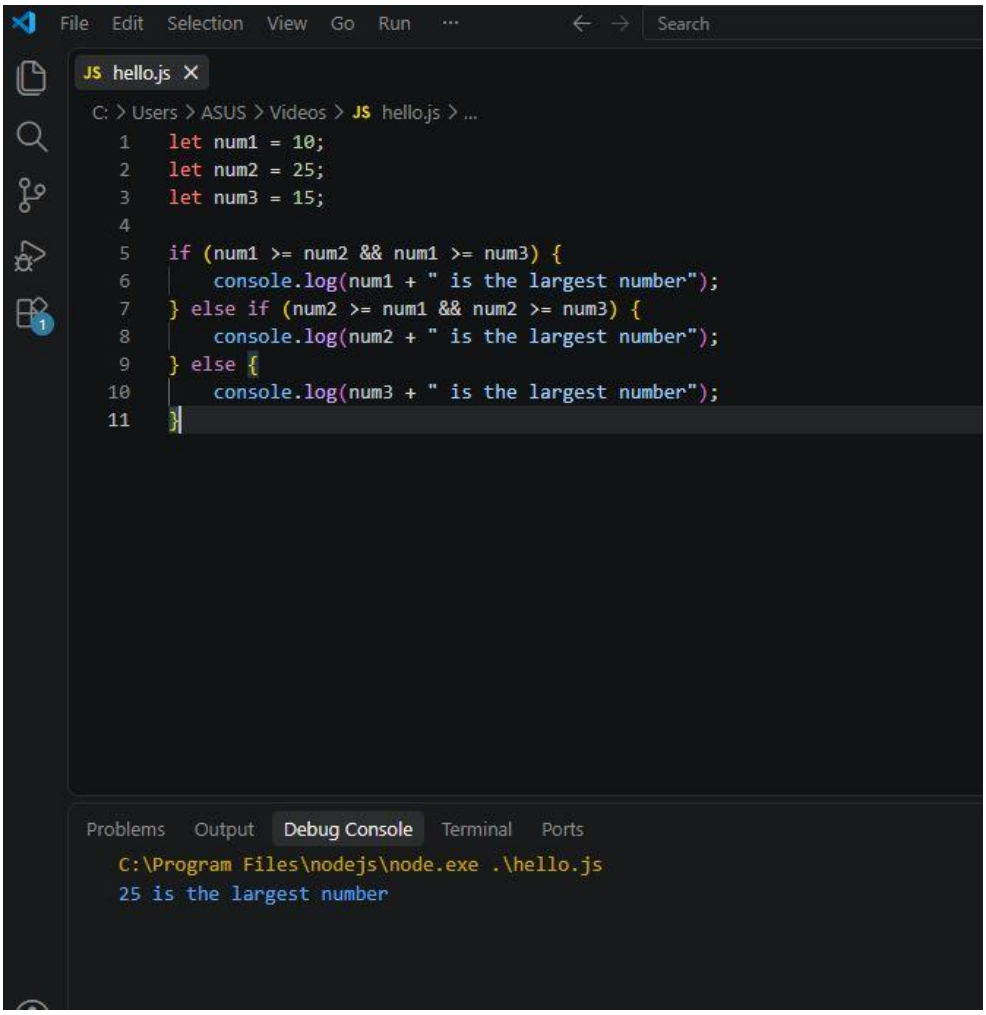
Write a JavaScript program to compare three numbers and find the largest one.

Source Code

```
let num1 = 10;
let num2 = 25;
let num3 = 15;
```

```
if (num1 >= num2 && num1 >= num3) {
    console.log(num1 + " is the largest number");
} else if (num2 >= num1 && num2 >= num3) {
    console.log(num2 + " is the largest number");
} else {
    console.log(num3 + " is the largest number");
}
```

Output



Output: 25 is the largest number.

Explanation

The program compares the three numbers using if, else if, and else. The number that is greater than or equal to the other two is printed as the largest.

3. Reverse a String

Problem Statement

Write a JavaScript program to reverse the characters of a string.

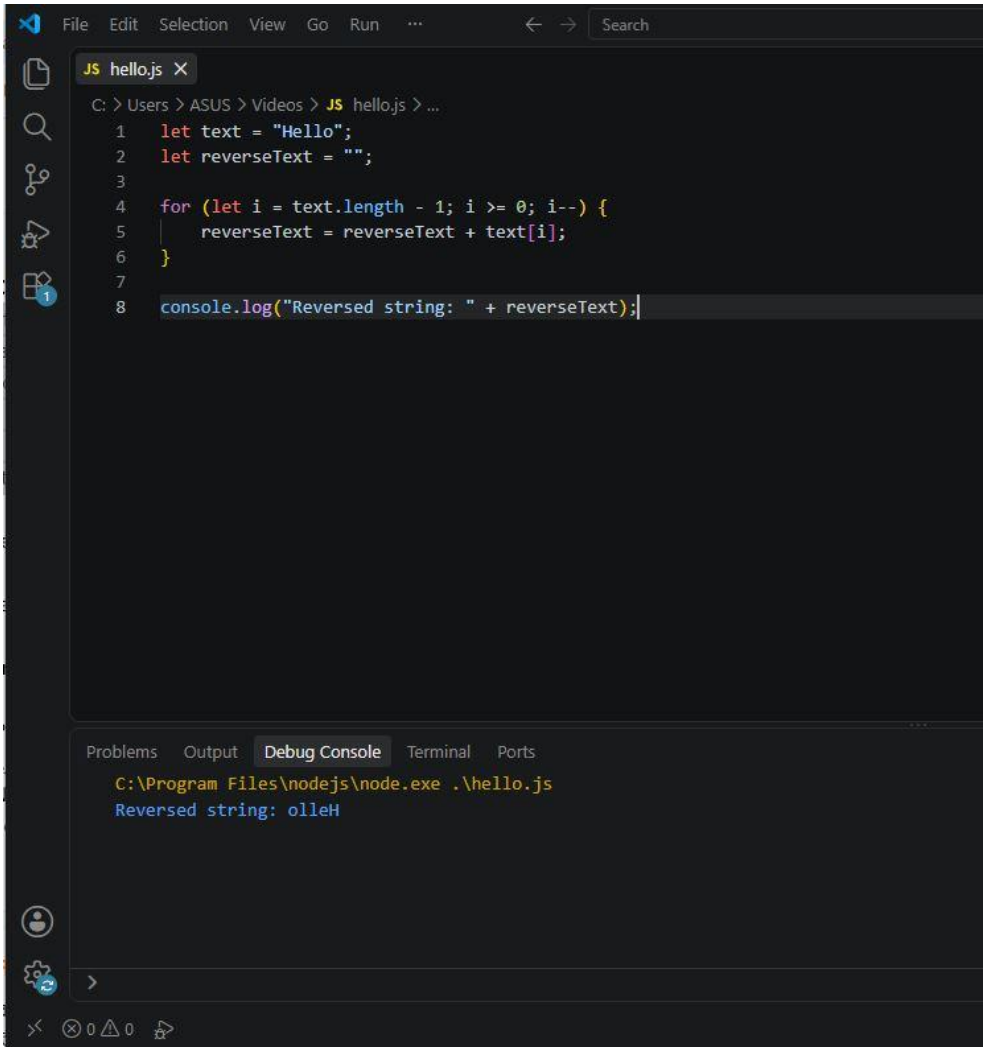
Source Code

```
let text = "Hello";
let reverseText = "";

for (let i = text.length - 1; i >= 0; i--) {
    reverseText = reverseText + text[i];
}

console.log("Reversed string: " + reverseText);
```

Output



Output: Reversed string: olleH.

Explanation

The loop starts from the last character of the string and moves backward. Each character is added to reverseText until the complete reversed string is created.

4. Count Vowels in a String

Problem Statement

Write a JavaScript program to count how many vowels are present in a string.

Source Code

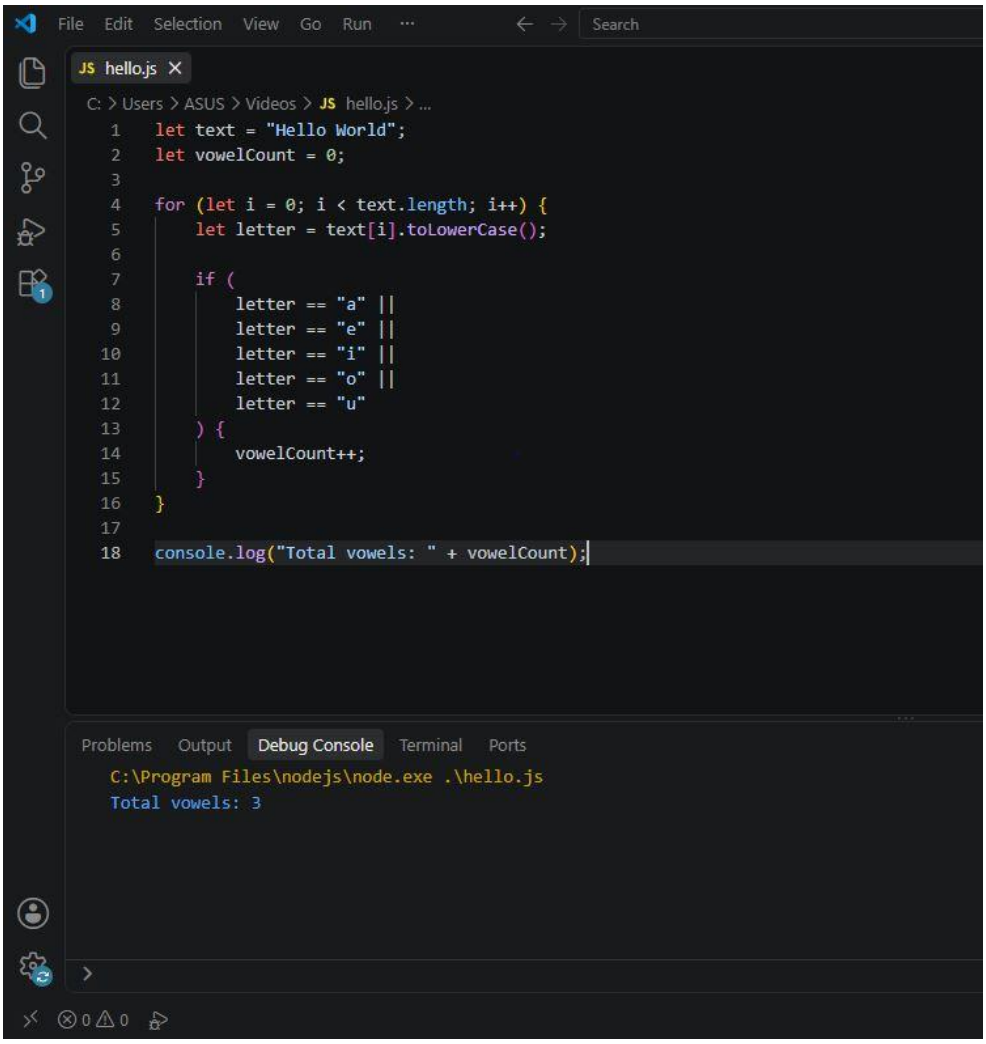
```
let text = "Hello World";
let vowelCount = 0;

for (let i = 0; i < text.length; i++) {
    let letter = text[i].toLowerCase();

    if (
        letter == "a" ||
        letter == "e" ||
        letter == "i" ||
        letter == "o" ||
        letter == "u"
    ) {
        vowelCount++;
    }
}

console.log("Total vowels: " + vowelCount);
```

Output



Output: Total vowels: 3.

Explanation

The program checks each character one by one. It converts the character to lowercase and compares it with a, e, i, o, and u. Every time a vowel is found, vowelCount increases by 1.

5. Remove Duplicate Values from an Array

Problem Statement

Write a JavaScript program to remove repeated values from an array.

Source Code

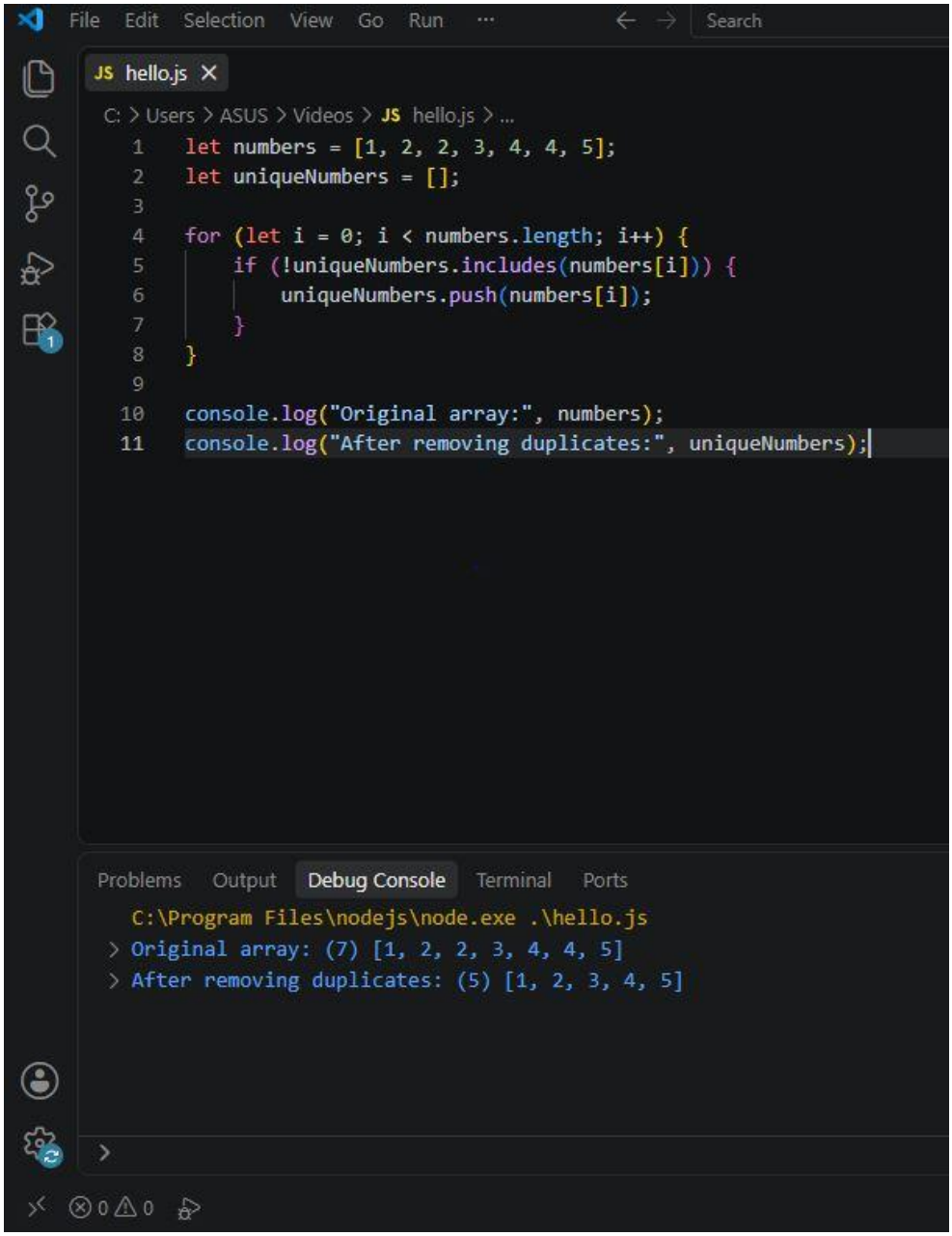
```
let numbers = [1, 2, 2, 3, 4, 4, 5];
let uniqueNumbers = [];

for (let i = 0; i < numbers.length; i++) {
    if (!uniqueNumbers.includes(numbers[i])) {
        uniqueNumbers.push(numbers[i]);
    }
```

```
}

console.log("Original array:", numbers);
console.log("After removing duplicates:", uniqueNumbers);
```

Output



Output: [1, 2, 3, 4, 5] after removing duplicates.

Explanation

The program goes through the original array and checks whether each value already exists in uniqueNumbers. If it does not exist, the value is added. This leaves only unique values in the new array.

Q2. Introduction to Node.js

1. What is Node.js?

Node.js is a JavaScript runtime environment that allows JavaScript code to run outside a web browser. It is commonly used to build server-side applications, command-line tools, APIs, and automation scripts. With Node.js, we can use the same JavaScript language for both browser-based work and many backend or testing tasks.

2. How does Node.js work?

Node.js uses the V8 JavaScript engine to execute JavaScript code. It follows an event-driven and non-blocking approach. Instead of waiting for one slow operation to finish before doing anything else, Node.js can continue handling other work and process the result when the operation is complete. This makes it useful for applications that deal with many requests or input/output operations.

3. Advantages of Node.js for QA Automation

1. JavaScript is widely used by modern web automation tools, so it is convenient for testers who already know JavaScript.

2. Node.js has a large package ecosystem through npm, which makes it easy to add testing and automation libraries.

3. It works well with popular automation frameworks such as Playwright, Cypress, and WebdriverIO.

4. Asynchronous programming is useful when tests need to wait for web pages, API responses, files, or other operations.

5. Node.js can run automation scripts from the terminal and can also be used in CI/CD pipelines.

4. Browser JavaScript vs Node.js

Topic	Browser JavaScript	Node.js
Runs in	Web browser	Computer/server environment
Main use	Web page interaction and front-end logic	Backend, scripts, APIs and automation
DOM access	Can access document and window	No browser DOM by default
File system	Restricted direct file-system access	Can work with files using Node.js APIs
Packages	Usually loaded/bundled for browser use	Can install packages using npm

Conclusion

This assignment helped me practice basic JavaScript concepts such as conditions, loops, strings, and arrays. It also introduced how Node.js allows JavaScript to run outside the browser and why it is useful for QA automation.